Workforce Gap Analysis

Digital Transformation Program | FY2025-FY2027

Prepared by: Lead People Analytics Manager
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1. Executive Summary

The single most critical finding is the Data Engineering gap. GlobalTech has 42 Data Engineers. Demand is 70 in Year 1 and 120 by Year 3, a gap of 28 now and 78 by FY2027. The scenario brief names this segment the highest priority because the core data platform depends on it. Annual attrition is 18 percent, rated Very High. The skills file shows a mobility pool of only 8. Cloud coverage is 2.8 and cybersecurity coverage is 2.5, against a Year 3 proficiency target of 5.

Supply in the six critical segments is 364. Year 3 demand is 765, a structural gap of 401. Year 1 demand is 523, a gap of 159. Cybersecurity must reach 60 by Q4 2024, from 31 today: a Year 1 gap of 29 and a Year 3 gap of 54. AI/ML attrition is 20 percent, with Very High scarcity, on a gap of 40. Digital Client Advisors are the largest gap, 140, and the best internal conversion, drawing on about 180 Traditional Advisors.

The plan closes all 401 roles. Internal moves cover 161 roles, 40.1 percent, which meets the CHRO mandate. External hiring covers 240. Fully loaded Year 1 hire costs, reskilling, retention, and vendor support total $47.79 million. Contingency of $2.21 million holds the plan at $50 million. Year 1 cash goes first to 70 Data Engineers and 60 Cybersecurity Specialists.

2. Current State Workforce Assessment

GlobalTech employs 15,000 people. The six critical segments are 364 people, 2.4 percent of the firm. Technical tenure is short: Data Engineers 3.2 years, Cloud/DevOps 2.9 years, and AI/ML Engineers 2.4 years. Digital Client Advisors average 6.8 years and RegTech Analysts 5.6 years.

Overall coverage is below 3.0 everywhere: Data Engineers 2.77, Cybersecurity 2.52, Digital Client Advisors 2.25, Cloud/DevOps 2.73, AI/ML 2.83, and RegTech 2.45. Scores from 2.0 to 2.9 are a significant gap. Core craft is stronger than those averages: data and analytics 4.2, cybersecurity 4.3, cloud 4.1, AI/ML 4.2, digital client 3.9, and RegTech 3.8. Adjacent skills are the problem. Digital client coverage is under 2.0 in the four engineering segments. AI/ML coverage for Digital Client Advisors is 1.4, below the severe-gap line of 1.5.

Feeder pools decide what can be grown inside. Traditional Advisors number 180, with high reskilling potential, digital client coverage of 2.9, and a mobility pool of 90. IT Infrastructure numbers 95, with cloud coverage of 3.8 and a pool of 15. Risk and Compliance numbers 112, with RegTech coverage of 4.1 and a pool of 28. Data Engineering, cybersecurity, and AI/ML pools are only 8, 5, and 3.

Portfolio attrition is 11.2 percent. Five segments sit above it: AI/ML 20 percent, Data Engineers 18, Cloud/DevOps 17, Cybersecurity 16, and Digital Client Advisors 13. RegTech is at 9 percent. The demand workbook shows Data Engineer cumulative attrition of 19 and rates Digital Client Advisors Medium. The attrition CSV, aligned to the scenario brief, shows cumulative attrition of 21 and a High rating. Its rule treats 12 to 17 percent as High. This analysis uses the CSV.

3. Future State Requirements by Segment

The forecast, and the gap versus today's supply, is as follows. Data Engineers: demand 70, 80, 120 and gaps 28, 38, 78, with Year 3 skill 5 and High scarcity. Cybersecurity: 60, 72, 85 and gaps 29, 41, 54, skill 5, High scarcity. Digital Client Advisors: 260, 310, 350 and gaps 50, 100, 140, skill 4, Medium scarcity. Cloud/DevOps: 55, 72, 90 and gaps 27, 44, 62, skill 5, High scarcity. AI/ML: 30, 43, 55 and gaps 15, 28, 40, skill 5, Very High scarcity. RegTech: 48, 57, 65 and gaps 10, 19, 27, skill 4, Medium scarcity.

Required headcount totals 523, 634, and 765. Gaps total 159, 270, and 401. Year 2 Data Engineer demand is 80, a gap of 38. That forecast figure is the one used here.

Year 1 is the platform and audit year: Data Engineering at 70, and Cybersecurity at 60 before Q4 2024. Year 2 is the cloud migration from Q1 2025, when Cloud/DevOps and AI/ML scale. Year 3 finishes digital advisory and RegTech automation. Hiring can be national. Technical reskilling needs six months, advisory reskilling about three. A cloud vendor and a cybersecurity MSSP can bridge Year 1 only. Closure is additive.

4. Quantified Gaps and Composite Risk

Year 3 headcount gaps are 78, 54, 140, 62, 40, and 27. Skill gap, required level minus overall coverage, is 2.23, 2.48, 1.75, 2.27, 2.17, and 1.55. All six overall scores are below 3.0, so every segment is flagged. Cybersecurity's skill gap is the widest, at 2.48.

The composite uses file inputs only. Gap magnitude is 40 percent: Year 3 gap divided by current headcount, scaled so a ratio of 3.0 would score 10. Attrition is 35 percent: the annual rate divided by 20 percent, times 10. Scarcity is 25 percent: Very High 10, High 8, Medium 5.

Scores are AI/ML Engineers 9.6, Cloud/DevOps 7.9, Data Engineers 7.6, Cybersecurity 7.1, Digital Client Advisors 4.4, and RegTech 3.8. AI/ML leads because 15 to 55 is a 267 percent increase, with 20 percent attrition and Very High scarcity.

Funding order applies the brief on top of that score. Data Engineers are the designated highest priority and have a Year 1 gate of 70, so they are rank 1. Cybersecurity has an audit gate of 60 by Q4 2024, so it is rank 2. AI/ML and Cloud/DevOps follow on the Year 2 migration. Digital advisory and RegTech are last: scarcity is Medium, and feeder pools can carry most of the volume. Scores near 7 and above are the critical band.

CSV cumulative attrition is 21, 13, 72, 12, 7, and 9 for Data Engineers, Cybersecurity, Digital Client Advisors, Cloud/DevOps, AI/ML, and RegTech. The 401 gap assumes today's headcount holds. Retention spend is what keeps it from growing.

5. Prioritized Closure Strategies

Data Engineers, rank 1. Hire 58 and reskill 20, using the pool of 8 and IT Infrastructure staff who score 3.8 on cloud. Year 1 needs 28 more people. Reskilling takes six months, so Year 1 is permanent hires plus a contractor bridge. Fully loaded cost is $180,000, or $10.44 million. Retain with a bonus, an equity-like grant, and a path to Staff Engineer.

Cybersecurity, rank 2. Hire 46 and develop 8. The pool is 5 and reskilling potential is Low. An MSSP covers the Q4 2024 audit while hires ramp. Cost is $165,000 each, $7.59 million, plus certification support, a band review, and a national search.

AI/ML Engineers, rank 3. The pool is 3 and potential is Low. Hire 37, including a university path, and convert 3 internals. Cost is $200,000, $7.40 million. Replacement cost is $300,000. Use grants, research time, and publication support so seven expected departures do not erase the build.

Cloud/DevOps, rank 4. Reskill 18 from IT Infrastructure and hire 44 at $160,000, $7.04 million. Place most external hires in Year 2. Fund cloud certification and a project rotation against 17 percent attrition.

Digital Client Advisors, rank 5. Convert 90 Traditional Advisors in three months and hire 50 at $95,000, $4.75 million. This segment carries the 40 percent internal rule. External hires need financial-services and digital-channel experience.

RegTech, rank 6. Redeploy 22 from Risk and Compliance and hire 5 at $110,000, $0.55 million. That pool already scores 4.1 on RegTech, and attrition is 9 percent. Train the tools. Skip a broad search.

6. Investment Recommendations

Closure is 161 internal roles and 240 external roles, a 40.1 percent internal share. The brief hinted at 40 percent of dollars on external hiring, 35 percent reskilling, 15 percent retention, and 10 percent contingency. A Data Engineer costs $180,000 and an advisory reskill about $8,000, so that dollar split would miss the Year 1 gates. This plan meets the headcount rule and spends where scarcity is real.

Segment totals, at the brief's fully loaded Year 1 cost: Data Engineers $13.24 million (external $10.44, reskill $0.50, retention $0.90, vendor $1.40). Cybersecurity $10.59 million (external $7.59, reskill $0.20, retention $0.70, MSSP $2.10). AI/ML $8.68 million (external $7.40, reskill $0.08, retention $1.20). Cloud/DevOps $8.49 million (external $7.04, reskill $0.45, retention $0.60, vendor $0.40). Digital Client Advisors $5.87 million (external $4.75, reskill $0.72, retention $0.40). RegTech $0.92 million (external $0.55, reskill $0.22, retention $0.15).

Program totals: external hiring $37.77 million, reskilling $2.17 million, retention $3.95 million, vendor support $3.90 million. Subtotal $47.79 million. Contingency $2.21 million. Grand total $50.00 million.

Year 1, about $17.4 million, buys the two gates, the bridges, and the first advisory cohort. Year 2, about $17.8 million, buys the migration and the AI/ML build. Year 3, about $12.6 million, finishes advisory and RegTech. Hold Year 2 until 70 Data Engineers and 60 Cybersecurity Specialists are on payroll. If Data Engineering attrition stays at 18 percent, move contingency into retention before opening more requisitions.

First dollars buy Data Engineering and Cybersecurity. Internal mobility then carries advisory and RegTech, which is what keeps the full portfolio inside $50 million.